Republic of the Philippines
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**College of Informatics and Computing Sciences**

**PROJECT PROPOSAL**
OOP 101 - OBJECT ORIENTED PROGRAMMING
Computer Science Department
1st Semester, AY 2026-2027

## I. TEAM INFORMATION

| | | | |
|---|---|---|---|
| **Team Name:** | | | |
| **Team Number:** | | **Section:** | IT - 2101 |

| **TEAM MEMBERS** | | | |
|---|---|---|---|
| **No.** | **SR-Code** | **Name** | **GitHub Username** |
| **1** | 25-06274 | Sion Luis E. Berania | Aoi77-ult |
| **2** | 25-07119 | Martha S. Gutierrez | |
| **3** | 25-00129 | Allianna H. Lucero | Ayxnna25 |
| **4** | 25-05200 | Darick Gaile A. Pascua | chrysaliafluerte-tech |

## II. PROJECT INFORMATION

**Project Title:** Supply Auditor for Pharmacies System

**Project Background**

Managing pharmacy inventory manually can sometimes cause problems, especially during tracking of medicine stocks, sale transactions, expiration dates and supply audits. Pharmacists need to keep records of supply sold to buyers, current stock quantities, batch details, and when products are nearing expiration. During manual recording, mistakes can occur such as incorrect stock availability, unrecorded sales, unnoticed expired medicines and generally the difficulty of inventory audits. The proposed Supply Auditor for Pharmacy System is a java-based program that will help pharmacists and inventory auditors organize medicine records, sale transactions, stock updates, expiration alerts and audit reports. The system will automatically update medicine availability whenever an item is sold or added and allow auditors to quickly identify the discrepancies between the physical counts and records. This application aims to make pharmacy auditing more efficient, organized and reduce the errors that may occur during manual auditing.

**Project Objectives**

To develop a Java-Based Supply Auditor for Pharmacy System that will assist in managing medicine inventories, sale transactions, stock expiration, and audit records while avoiding errors in manual pharmacy recording.

**General Objectives**

To develop a Java-based application that will help automate pharmacy inventory management, track sale transactions, monitor expiration dates, and streamline audit reconciliations while reducing errors.

**Specific Objectives**

The project aims to:

1. Implement a secure login system with role-based access for pharmacists and inventory auditors.
2. Track medicine stock quantities, batch details, sale transactions, and expiration dates in real time.

3. Automate audit reconciliations by comparing physical stock counts with system records to detect discrepancies.
4. Generate clear reports for sale histories, low stock alerts, product expiration, and audit summaries.

**Target Users**

| Target User | How They Will Use the System |
|---|---|
| Pharmacist/ Store Manager | Records daily sale transactions when customers purchase medicine, updates stock details, checks expiring items, and manages medicine inventory entries. |
| Inventory Auditor | Conducts periodic stock audits, inputs physical shelf counts, checks generated discrepancy reports and verifies audit history |

**Proposed Features**

| Feature | Description |
|---|---|
| User Authentication & Role Management | Provides secure login screens with different permissions for administrative, auditing, and pharmacist access. |
| Inventory and Batch Management | Stores complete details of medicine, including product names, categories, prices, batch numbers, and expiration dates. |
| Sales and Dispensing Tracker | Allows pharmacists to log sold medicines upon customer purchase, automatically deducting sold quantities from current stock. |
| Automated Audit and Reconciliation | Enables auditors to input stock quantities, calculates the variances against system records and highlights discrepancies. |
| Expiry and Low Stock Alerts | Shows notifications and highlighted list of medicines that are nearing expiration or are running low. |
| Reporting and Export | Generates the summary reports for daily sales, inventory balances and audit record with options to export |

## III. JAVA AND OOP CONCEPTS

| OOP Concept | How It Will Be Used |
|---|---|
| Classes and Objects | Used for creating entity templates like; **Product, User, SaleRecord, AuditRecord** and **InventoryBatch.** |
| Methods | Encapsulated system tasks such as; **deductStockOnSale(), calculateDiscrepancy(), checkExpiration()** |

| | |
|---|---|
| | and **generateAuditReport().** |
| Encapsulation | Protects product data and user credentials by using private variables and public getter/setter methods |
| Inheritance | Uses a base **User** class extended by child classes like **Pharmacist** and **Auditor** to reuse common properties. |
| Polymorphism | Allows different log entries or notifications to be formatted and handled through shared method interfaces |
| Abstraction | Defines clear guidelines for reporting and auditing functions using abstract classes or interfaces without showing the back-end logiv |
| Exception Handling | Prevents system crashes during wrong user input or file loading errors using **try-catch** blocks. |
| Arrays/Data Handling | Uses Java collections such as **ArrayList** and **Hashmap** to hold and search through list of productions, sales records and audit logs. |

## IV. PROPOSED CLASSES

| Class | Purpose | Main Attributes | Main Methods |
|---|---|---|---|
| Product | Represents the medicinal item in the pharmacy stock. | productID, name, category, unitPrice, stockQuantity, expirationDate | updateStock(), isExpired(), getDetails() |
| InventoryBatch | Tracks batches of medicines with expiration dates. | batchID, productID, quantity, manufactureDate, expiryDate | validateBatch(), getRemainingDays() |
| SaleRecord | Stores details of medicine sold to customers by pharmacists. | transactionID, productID, quantitySold, totalPrice, saleDate, pharmacistId | processSales(), deductStock(), getReceiptDetails() |
| Auditor | Handles physical stock checking and audit tasks | auditorID, name, assignedSection, auditLogs | conductAudit(), generateDiscrepancy Report() |
| AuditRecord | Saves the summary and findings of an audit session | auditID, date, auditorID, discrepancyList, status | saleRecord(), exportReport() |

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## V. PROGRAM FLOW

Start
↓
Main Menu
↓
Select Operation
↓
Enter Information
↓
Validates Input
↓
Process Request
↓
Display Result
↓
Return to Main Menu
↓
Exit

## VI. TEAM RESPONSIBILITIES

| Member | Assigned Tasks |
|---|---|
| Sion Luis E. Berania | Lead Programmer |
| Martha S. Gutierrez | Documentation |
| Allianna H. Lucero | System and Design Leader |
| Darick Gaile A. Pascua | Project Leader/ Lead Programmer |

## VII. DEVELOPMENT SCHEDULE

| Week | Activity | Expected Output |
|---|---|---|
| Week 1 | Planning and requirements | Project Proposal |
| Week 2 | System and class design | Class/UML diagram |
| Week 3 | Initial coding | Basic classes |
| Week 4-5 | Feature development | Working features |
| Week 6 | OOP implementation | Completed OOP features |
| Week 7 | Testing and debugging | Tested application |
| Week 8 | Documentation and presentation | Final project |

## VIII. EXPECTED OUTPUT

The team expects to produce a functional java application that manages pharmacy inventory, tracks sales transactions, monitors stock expiration dates and automates audit reconciliations with discrepancy reports. The application will demonstrate the appropriate use of java and object oriented programming concepts.

## IX. PROJECT LIMITATIONS

- The application runs locally on a desktop and does not include cloud synchronization between branches.
- Input is handled manually using standard id entry.

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## X. GITHUB REPOSITORY

Supply-Auditor-Pharmacies-System
https://github.com/chrysaliafluerte-tech/Supply-Auditor-Pharmacies-System/tree/main

## XI. APPROVAL

**Prepared by:**

| **Name** | **Signature** |
|---|---|
| Sion Luis E. Berania | |
| Martha S. Gutierrez | |
| Allianna H. Lucero | |
| Darick Gaile A. Pascua | |

**Instructor's Comments:**

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**Project Status:**

- Approved
- Approved with Revisions
- Revise and Resubmit

Name of Instructor:
**Mr. Jayson Abratique**

**Date Signed:** _____